Social Network Analysis

LINK ANALYSIS

SLIDE CREDITS: BOOK BY PROF. TANMOY CHAKRABORTY

What are the Links?

- Model of interaction between entities defines types of entities being connected and types of links that connect these entities
- Diversities in connected entities
  - Homogeneous versus heterogeneous
- Diversities in connecting links
  - Directed versus undirected
  - Weighted versus unweighted
  - Signed versus unsigned, etc.
- Dynamics of link formation yields formation of substructures in the network
  - Communities emerges due to homophily
  - Strong ties and weak ties, etc.

Why Link Analysis?

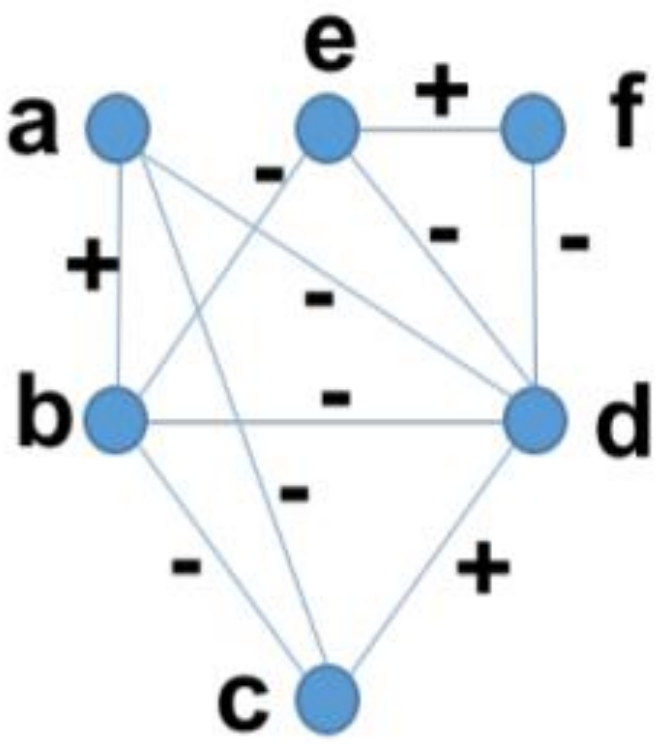
Fundamental output of link analysis task is to perform link-based object ranking, using global (network-wide) metric to measure the comparative importance of a node in the network.

- Entity Ranking
  - Search Engine Optimization
  - Scientific article Ranking
  - Scientific Author Ranking, etc.

Why Link Analysis?

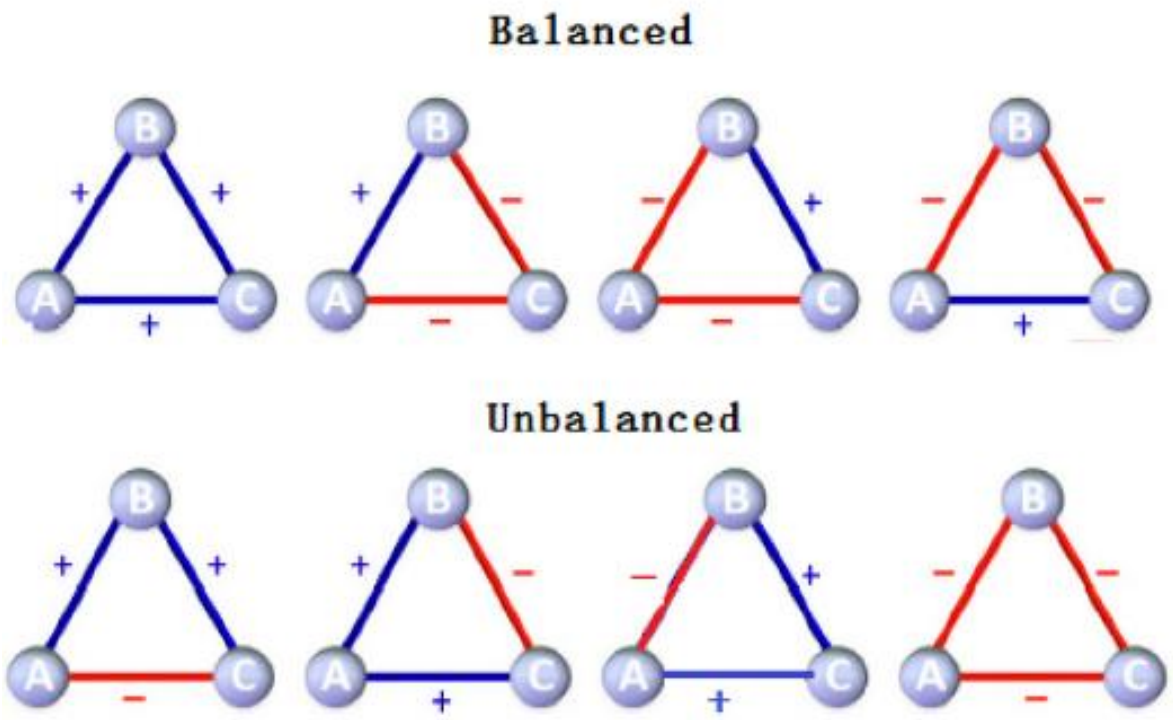
- Anomaly Detection
  - Online Fraud Detection
  - Counter Terrorism
  - Police/Military intelligence, etc.
- Adversarial Attacks
  - Adversarial attack on selected nodes to disrupt services
- Mining New Patterns
  - Crime Prevention
  - Future rank prediction
  - Link Prediction
  - Market Research, etc.

Signed Networks



- Direction of a link in a network captures the direction of information flow across the link
- Weight of a link in a network represents the strength of influence of information passing through that link
- Neither of the above express how the information is perceived by the receiving node!
- There often exist element pairs in perception/reaction towards information content –
  - ✓ like/dislike (YouTube),
  - ✓ agree/disagree (Reddit),
  - ✓ Positive review/negative review (Amazon), etc.
- Signed network captures the above opinion/relationship dynamics across entities

Balance Theory: Triads



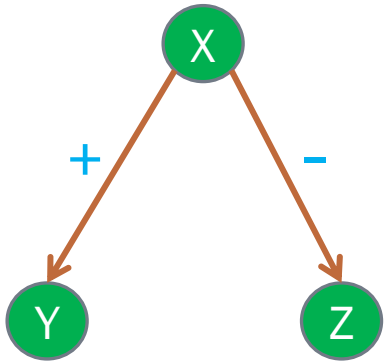
Positive = Friendship, Negative = Enmity

Li and Tang 2012

- Balance state occurs in triads when all sign multiplication of its sentiment relation charges positive

- Three Positive links
  - mutual trust and respect
  - Stable
- Two negative, one positive
  - trust between friends established based on distrust towards a common enemy
  - Stable
- Two positive, one negative
  - mutual friends would be under stress to take sides
  - Unstable
- Three negative links
  - No mutual trust
  - Unstable and likely to be disintegrated

Signed Networks: Status Theory

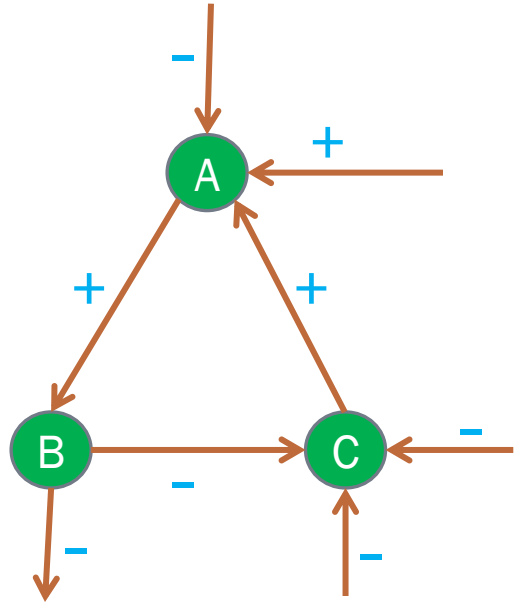


$Y > X > Z$

Status relative to X

- Balance theory views signed links as model of likes and dislikes
- a signed link formed can have other possible interpretation!
  - Interpretation of link-sign as an indicator of relative status/prestige of a node with respect to the other
  - Status Theory
  - Assumes a signed, directed network of the entities
- A initiates a positive link to $B \Longrightarrow A$ considers B to have a higher status than itself
- A initiates a negative link to $B \Longrightarrow A$ considers B to have a lower status than itself

Signed Networks: Status Theory



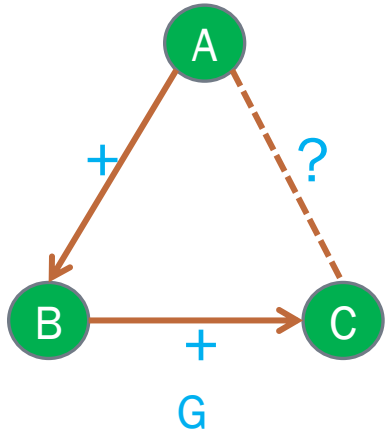
Snapshot of a signed graph

- Node-level metrics defined in this connection:
  - Generative Baseline (g): The fraction of positive signs generated by a node
  - Receptive Baseline (r): The fraction of positive signs received by a node

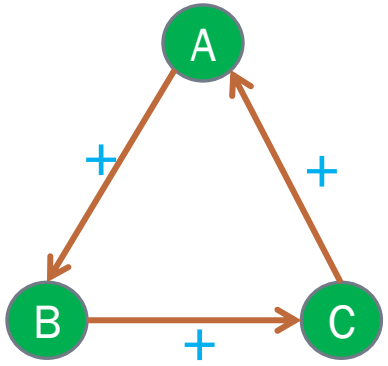
- Scores for generative baselines of the nodes of the signed graph beside are as follows:
  - $A_g = \frac{1}{1} = 1,$ $B_g = \frac{0}{2} = 0,$ $C_g = \frac{1}{1} = 1$

- Scores for receptive baselines of the nodes of the signed graph beside are as follows:
  - $A_r = \frac{2}{3} = 0.67,$ $B_r = \frac{1}{1} = 1,$ $C_g = \frac{0}{3} = 0$

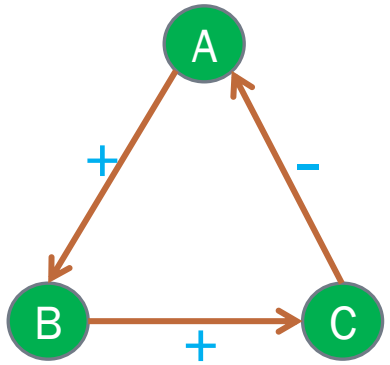
Comparison: Balance Theory and Status Theory



- Theory of status makes sense for directed networks only
- Theory of balance, though originated for undirected graphs, are also applicable for directed graphs
- In directed network G, if C forms a link to A, which link-sign is most likely to occur for that link?
  - According to theory of balance, link CA is predicted to be a positive link
  - According to theory of status, link CA is predicted to be a negative link!
- The two theories may infer conflicting predictions, as they have different interpretations altogether

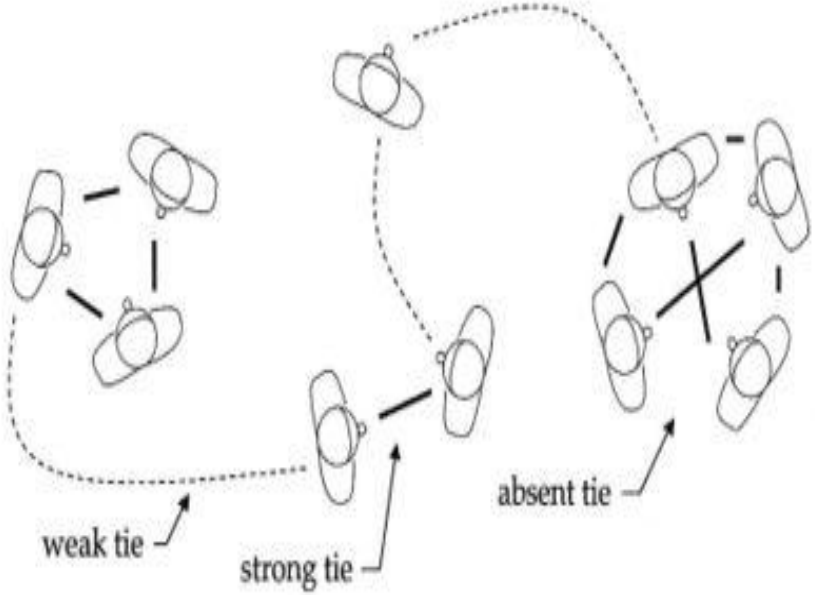


Balance Theory



Status Theory

Interpersonal ties



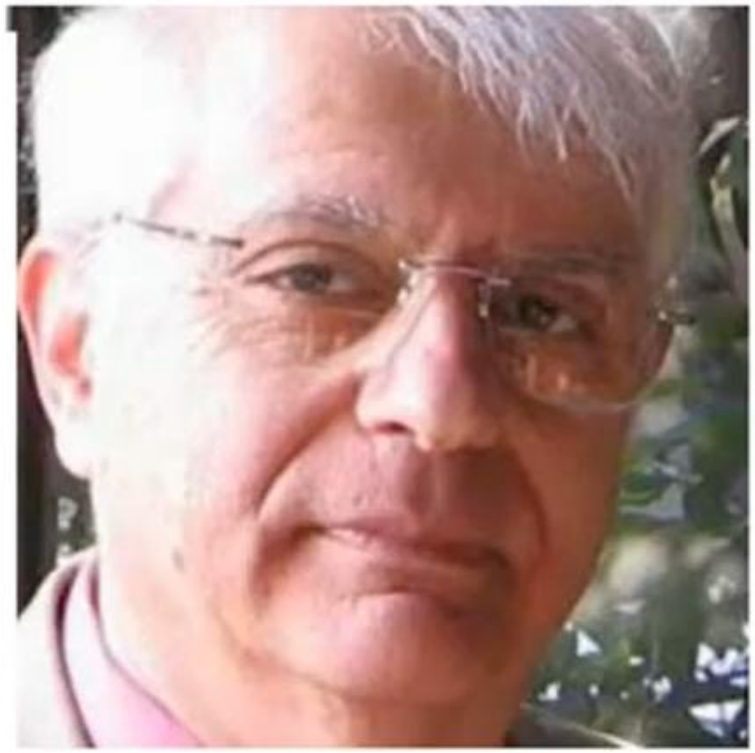
https://en.wikipedia.org/wiki/Interpersonal_ties

- Defined as information-carrying connections between entities/people
- Appear generally in three varieties: strong, weak or absent
- Strong ties
  - develop among entities that share interest and beliefs
  - thought of as source of confidence and emotional dependency
- Weak ties are mere acquaintances

The Strength of Weak Ties[1]

Mark S. Granovetter
Johns Hopkins University

Analysis of social networks is suggested as a tool for linking micro and macro levels of sociological theory. The procedure is illustrated by elaboration of the macro implications of one aspect of small-scale interaction: the strength of dyadic ties. It is argued that the degree of overlap of two individuals' friendship networks varies directly with the strength of their tie to one another. The impact of this principle on diffusion of influence and information, mobility opportunity, and community organization is explored. Stress is laid on the cohesive power of weak ties. Most network models deal, implicitly, with strong ties, thus confining their applicability to small, well-defined groups. Emphasis on weak ties lends itself to discussion of relations *between* groups and to analysis of segments of social structure not easily defined in terms of primary groups.



Mark Granovetter

Link Prediction approach 1: Strength of a Tie

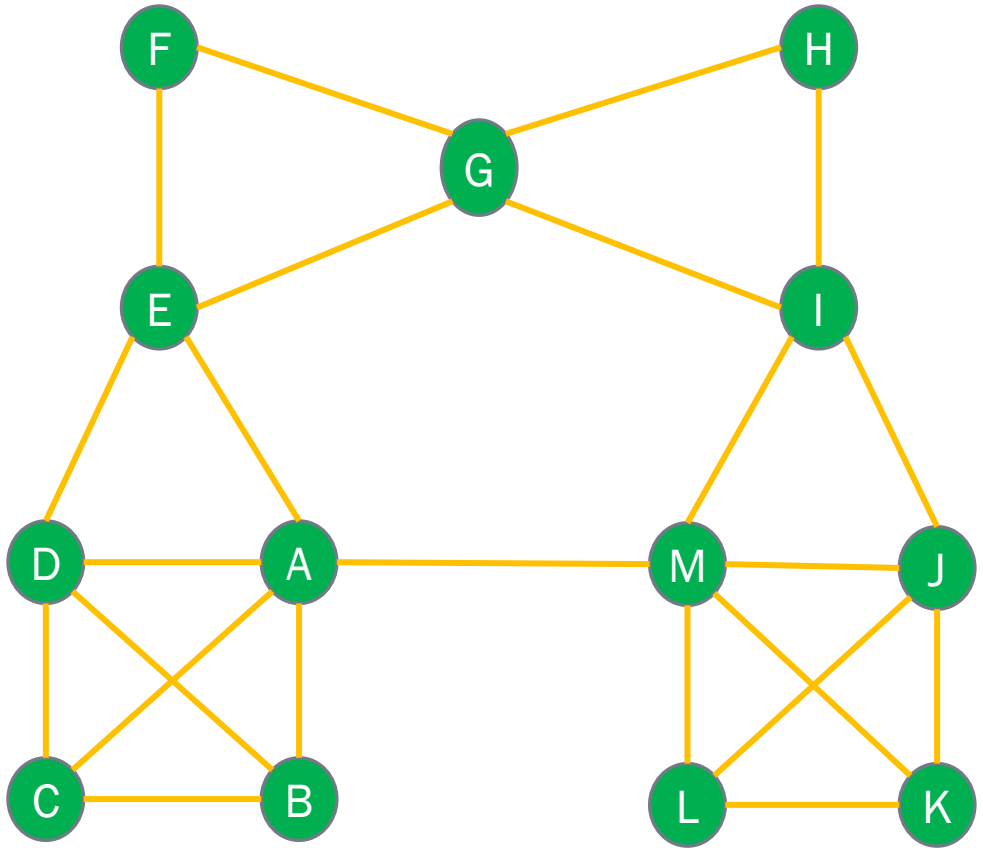
- Strength of ties captures a sense of closeness among entities/people
- Simplest metric to capture the same is via Jaccard score
- Corresponding metric, called Neighborhood Overlap (NO) is defined as:

$$NO(x, y) = \frac{|\Gamma(x) \cap \Gamma(y)|}{|\Gamma(x) \cup \Gamma(y)|}$$

where $\Gamma(\cdot)$ denotes the neighbourhood of a node

- Higher the $NO(\cdot)$ score, higher the overlap between the nodes, and higher the chance forming a link in between

Link Prediction approach 1: Strength of a Tie ; Neighborhood Overlap: Example



$\Gamma(A) = \{B, C, D, E, M\}$,
$\Gamma(M) = \{A, I, J, K, L\}$,
$\Gamma(E) = \{A, D, F, G\}$

$|\Gamma(A) \cap \Gamma(M)| = |\phi| = 0$
$|\Gamma(A) \cap \Gamma(E)| = |\{D\}| = 1$

$|\Gamma(A) \cup \Gamma(M)| = |\{B, C, D, E, I, J, K, L\}| = 8$
$|\Gamma(A) \cup \Gamma(E)| = |\{B, C, D, F, G, M\}| = 6$

$NO(A, M) = \frac{0}{8} = 0$, $NO(A, E) = \frac{1}{6}$

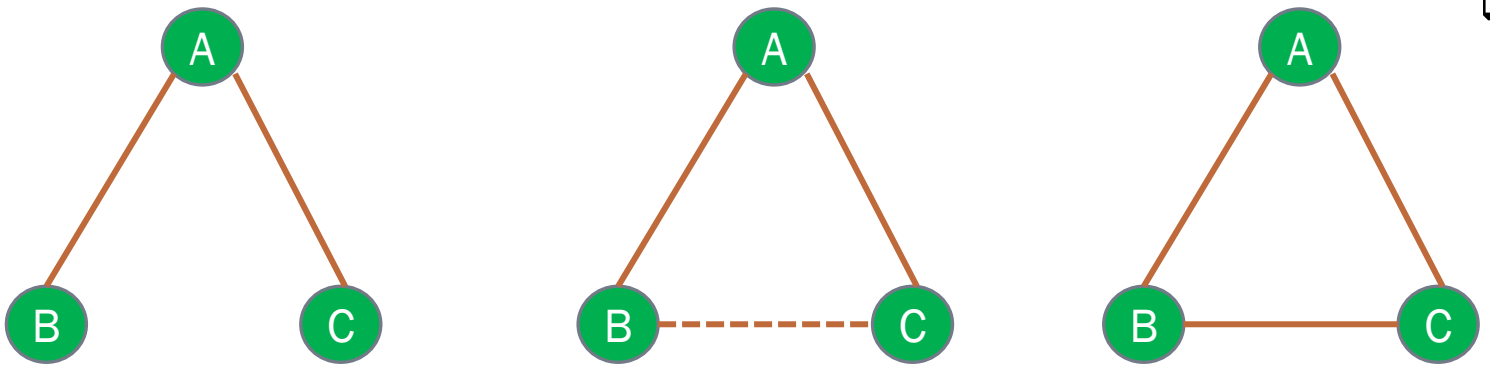
Dunbar Number

- An empirical study that supports the presence of strong and weak ties in real world
- Refers to a suggested cognitive limit to the number of people with whom one can maintain stable social relationships

  Empirically, this number ranges from 100 to 250. For general purposes this number is set at 150.
- First proposed in 1990 by British anthropologist Robin Dunbar
- Observed a correlation between primate brain size and average social group size
- Extrapolation from the results of primates using the average human brain size
- The number informally represents the set of people one can be in close contact with (strong ties)
- Rest of the social contacts are likely to be acquaintances (weak ties)

Link Prediction approach 2: Triadic Closure

- A friend of a friend is also a friend – is the philosophy



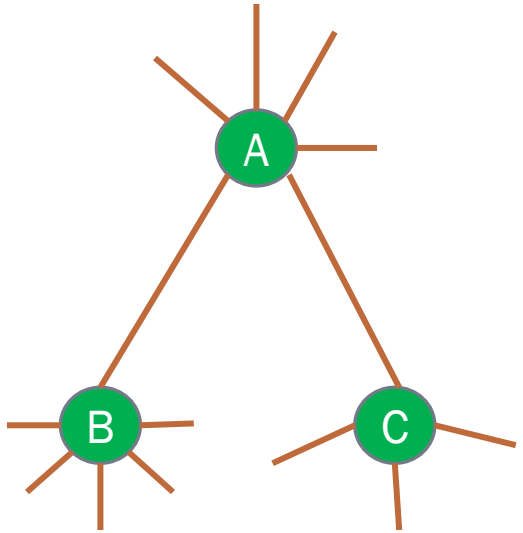
$t - \Delta t$
B & C are not friends yet

t
B & C gets introduced (via A)

$t + \Delta t$
B & C are friends now

- Reasons behind Triadic closure formation
  - Opportunity: of meeting via mutual connection
  - Trust: link formation based on mutual trust
  - Incentive: nodes may have incentives to bring their mutual friends together

Link Prediction approach 2: Triadic Closure…

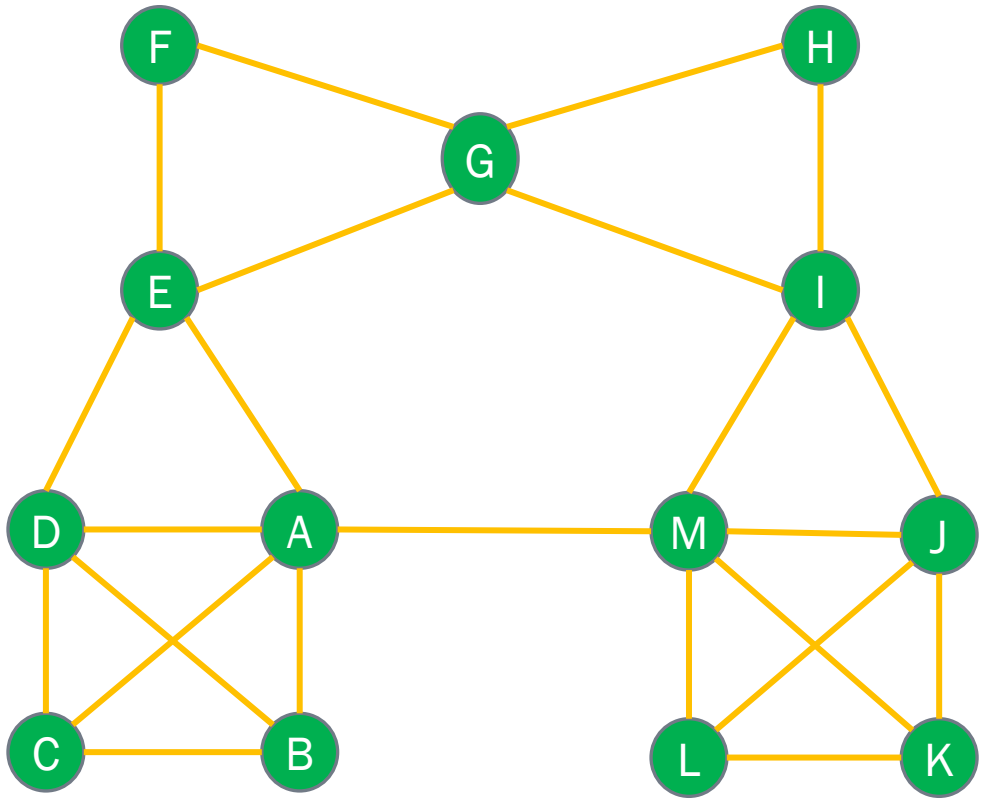


- Quantifying Strength of Triadic Closures
- Strength of a triadic closure with respect to node A and the nodes B and C of which A is a mutual friend can be quantified using the clustering coefficient of node A
- Clustering coefficient of a node (CC_A) measures the probability that the pair of friends (B and C) of the given node (A) are friends of each other

$$CC_A = \frac{2 \times \sum_{i,j \in \Gamma(A)} \boldsymbol{I}\big((i,j) \in E\big)}{k_A(k_A - 1)}$$

where $\boldsymbol{I}(\cdot)$ is the indicator function that returns 1 if condition is true, and 0, otherwise

Link Prediction approach 2: Triadic Closure - Example



- B and M are neighbours of node A. To find the how likely they form a link.

$\Gamma(A) = \{B, C, D, E, M\}$

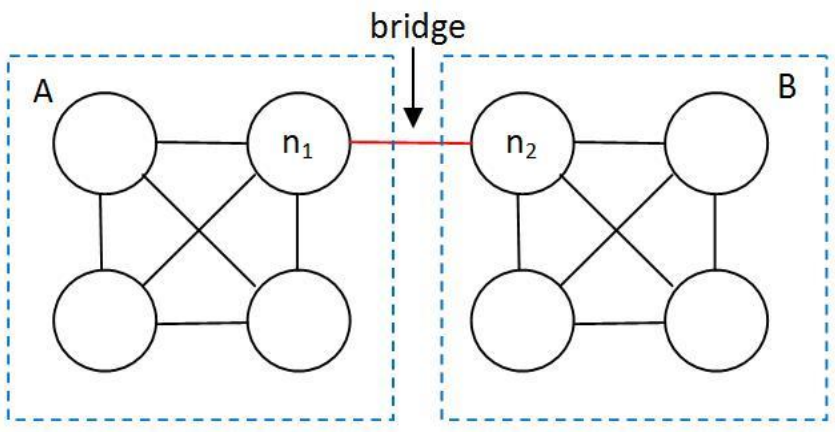
$k_A = 5$

Existing valid edges in $\Gamma(A)$ are $\{BC, BD, CD, DE\}$

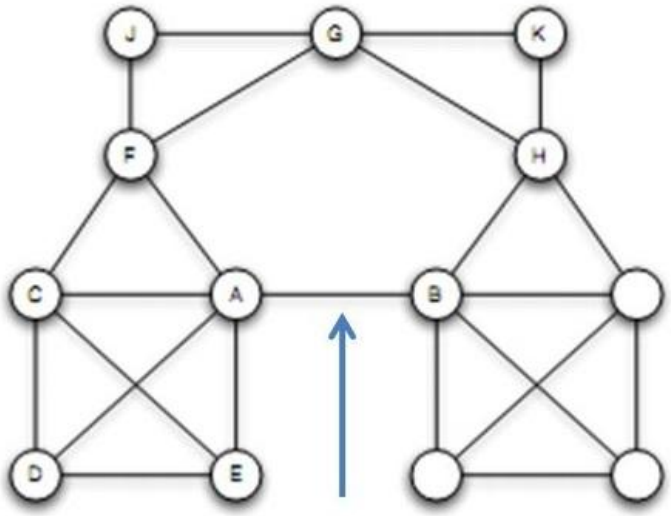
$CC_A = \frac{2\times4}{5\times4} = 0.4$

With 40% probability we may say that nodes B and M will form a link in the future.

Bridges and Local Bridges



https://en.wikipedia.org/wiki/Bridge_(interpersonal)



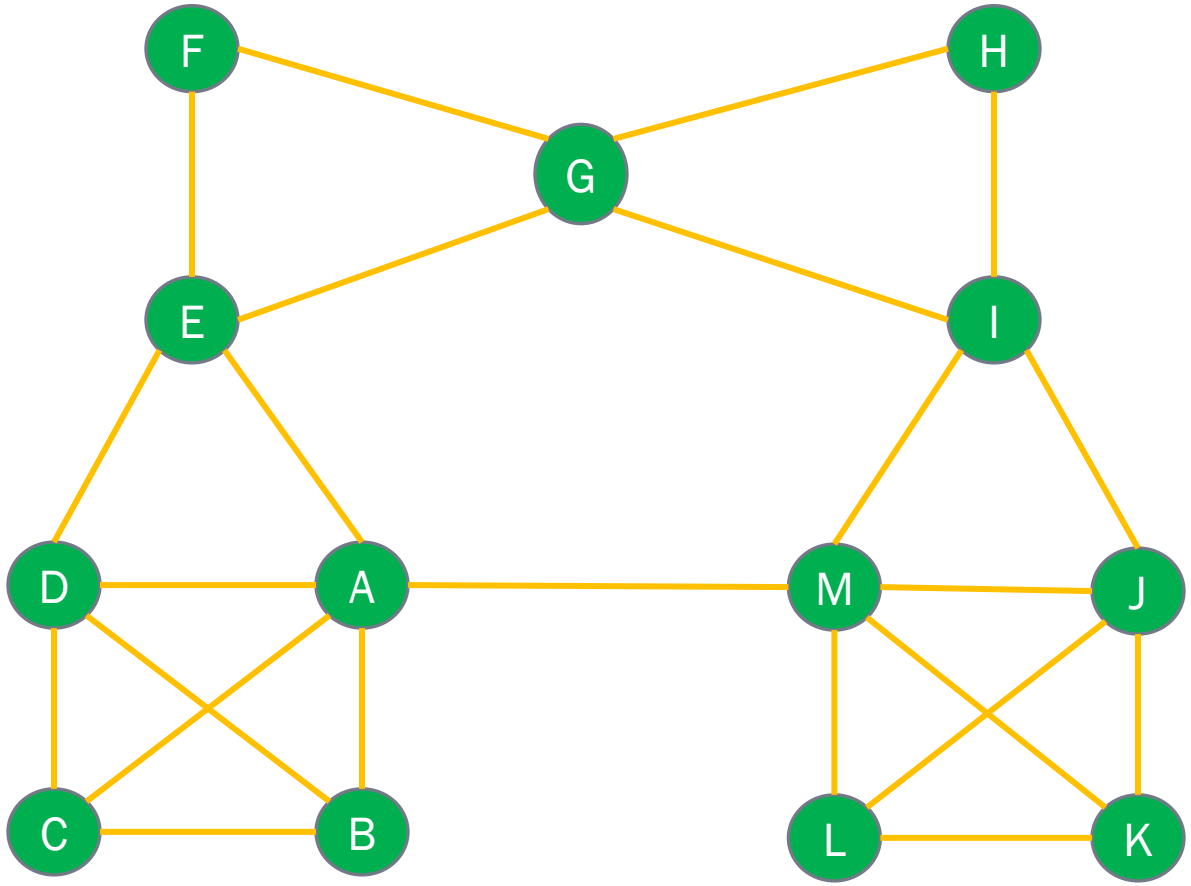
Local Bridge

https://slideplayer.com/slide/9361256/

- A bridge is a direct tie between nodes that would otherwise be in disconnected components of the graph
- Removal of a bridge increases the number of disconnected components in a network

- Local bridges are ties between two nodes in a social graph that are the shortest route by which information might travel from those connected to one end to those connected to the other
- On removal of a local bridge the distance between these two nodes will be increased to a value strictly more than two

Local Bridges/Weak Ties



- An edge can be considered a local bridge if its Neighborhood Overlap Score (NO) is zero
- In other words, end-points of a local bridge have no mutual friends
- Local bridges are not a part of any triad in the network
- (A, M) is a local bridge/weak tie

Local Bridges: Edge Embeddedness

- For an edge $\langle x, y \rangle$, its embeddedness can be defined as the number of mutual friends that the endpoints of the edge posses

$$Embeddedness(\langle x, y \rangle) = |\Gamma(x) \cap \Gamma(y)|$$

- A local bridge is an edge with embeddedness of zero

Local Bridges: Importance

- Close friends tend to move in the same circles that we do
  - Information close friends receive overlaps considerably
- Acquaintances, by contrast, know people that we do not,
  - People receive more novel information through acquaintances than from close friends
- Weaker ties act as a bridge and help a person gain access to newer and wider information (strength of weak ties)
- In case of stress/conflict between two groups, weak ties act as mediators
- In an adversarial setting, removing local bridges can lead to the formation of echo chambers
- During disease outbreaks, local bridges may cause the disease to transmit from one group to another